Budget Justification

Draft basis. The amounts below match the companion formula-driven workbook and total $486,436 over three years. Salary, fringe, and F&A values are planning assumptions and must be replaced with Georgia Tech's approved rates before submission. No voluntary committed cost sharing is proposed.

A. Senior Personnel

PI Brian Kyanjo: 7.2 person-months (60% of a 12-month appointment) in each project year. The PI will direct the project and lead model-adapter and PISM development, ICESEE/CryoBench integration, secure execution, releases, operations, and technical documentation. The workbook assumes a Year-1 base salary of $85,000 with 3% annual escalation, yielding $51,000, $52,530, and $54,106 in requested salary.

Co-PI Alexander A. Robel: one summer month in each project year. The Co-PI will lead science requirements and validation, the observational and cross-model demonstrations, user evaluation, and community engagement. The workbook assumes a nine-month academic salary of $130,000 with 3% annual escalation, yielding $14,444, $14,878, and $15,324.

Total senior-personnel salary is $202,282 over three years. Final person-months and institutional base salaries will be certified by Georgia Tech.

B. Other Personnel and C. Fringe Benefits

No postdoctoral scholar, student, or separate research software engineer is included in the current planning budget. The PI provides the principal software-engineering effort. Fringe benefits are estimated at 30% of salary, or $60,685 over three years. The final budget will use the applicable Georgia Tech fringe rates by appointment type. If a CI-professional position is added, the budget and required CI Professional Development Plan will be revised together.

D. Equipment

No equipment is requested. Computing will use existing Georgia Tech PACE and web infrastructure, user-managed resources where appropriate, and budgeted cloud services.

E. Travel

$14,400 is requested across three years ($4,800 per year). Each year includes one trip near NSF for the required CSSI PI meeting, estimated at $1,800, and $3,000 for presentation, workshop delivery, or stakeholder engagement at a cryosphere, PISM, or science-gateway venue. Costs cover economy airfare or ground transportation, lodging, per diem, and registration in accordance with Georgia Tech travel policy.

F. Participant Support Costs

$18,000 is requested ($6,000 per annual workshop) for travel, lodging, registration, or stipends for external participants. Selection will prioritize students, early-career researchers, minority-serving institutions, EPSCoR jurisdictions, and institutions without dedicated HPC/software staff. These funds are excluded from the MTDC base and will not be rebudgeted to other categories without required NSF approval.

G. Other Direct Costs

$21,000 is requested ($7,000 per year): $1,500 for materials and supplies supporting workshops, software testing, and documentation; $1,500 for open-access publication and archival dissemination; and $4,000 for bounded AWS compute/storage/network charges used to qualify and demonstrate the cloud backend. Cloud usage will be protected by quotas, budgets, and alerts. The bounded computing-access pilot described in the Project Description (planning scenario of about $1,560: 50 participants at $25 with 25% contingency) is not included in these totals; it requires institutional approval and cost classification before it is added. Development of the allocation service (Task 2.4) is part of the PI effort above; administering allocations and reconciling charges add operational effort that is not separately budgeted. Final classification of publication and cloud charges will follow institutional and NSF policy.

I. Indirect Costs and Cap Check

The planning workbook applies a provisional 57% F&A rate to Modified Total Direct Costs, excluding participant support, producing estimated indirect costs of $170,069. The final submission will use Georgia Tech's current federally negotiated rate and correct base. Estimated annual totals are $158,098, $162,105, and $166,233; the three-year total of $486,436 is $113,564 below the $600,000 Elements cap. Any personnel, subaward, rate, or scope change will be reconciled across the Research.gov forms, workbook, budget justification, management plan, and CI-professional plan.